PUV Pipeline: Methods and Validation Results

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How to read this document

This is a working draft of the methods and results sections written for the first PUV paper. It is kept because it is the most complete written account of the L1 to L3 processing and of the validation tests run in spring and summer 2026.

- **Methods** (Section 1) were checked against the code on 2026-10-15 and describe the processing that produced catalog version 2026-10-14.1. Each paragraph that changed carries a boxed note saying what changed and when.
- **Results** (Section 2) record analyses run between April and July 2026 on the catalog as it then stood (33 to 38 instruments, 17-minute segments for the earliest). None was recomputed for this revision. Passages later retracted are boxed and labeled.

For the current pipeline, start with `README.md`, then `docs/pipeline_levels.md` (status by level), `PIPELINE_NOTES.md` (design reasons) and `CHANGELOG.md`. The long-form manual is `docs/PUV_Pipeline_Guide.pdf`. The spectral-estimator choice is documented in `docs/multitaper_writeup.pdf`, and the faults of the legacy processing code in `docs/legacy_pipeline_defects.pdf`. Current model-comparison findings live in the separate paper repository (`Wave_model_validation_paper/docs/`), not here.

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1 Methods

1.1 Nearshore wave measurements

Waves and currents were measured with bottom-mounted Nortek Vector acoustic Doppler velocimeters (ADVs) recording pressure and velocity (PUV) continuously at 2 Hz. The catalog holds 65 instrument-records from 46 deployments on the San Diego County coast and at Catalina Island, 2014–2026, at nominal depths of 5–30 m (median 7.5 m over the 72 configured records). Deployment metadata (instrument heading, clock drift, surveyed position, and the pressure-port height above the bed, z_s, called `doffp` in the code) come from the field deployment notes. Across the 72 configured records z_s ranges from 0.38–1.09 m.

Changed 2026-10. The April draft described the TBR23 array only (nominal depths 5–7 m, z_s = 0.75–0.77 m). The ranges above are read from the deployment configs through `config/deployment_registry.m`.

1.2 Level 1: Quality control and coordinate rotation

Raw data are read either from the Nortek ExploreV ASCII export (`.dat`/`.sen`/`.hdr`) or from the recorder binary (`.VEC`) by `L1_raw_to_qc/read_VEC.m`. Each recorder file is treated as one burst. Clock drift measured at recovery is removed linearly over the deployment. For the pre-2019 recorders, whose internal clock ran from a wrong epoch, time is recovered from the `MMDDHHMM` file names; that clock was Pacific local time, so a fixed `instr.clockOffsetHours` is added to reach UTC (7 h for the 2014 records, 8 h for the later ones; each checked against the tide with `scripts/audit_clock_lag.m`). A gap between consecutive sensor records longer than `cutoffGapSec` (1 s by default, 60 s for the pre-2019 recorders) marks battery depletion, and all data after the first such gap are discarded. The gap test runs within each burst, so a seam between files is not read as a cutoff.

Quality control is applied per channel (`L1_raw_to_qc/puv_channel_qc.m`, thresholds from `l1_qc_opts.m`), so a failed thermistor or pressure sensor does not discard good velocity:

- *Velocity* fails where the minimum beam correlation is below 70%, where the frame is toppled (|pitch| or |roll| $\geq 30°$), or where the tilt is unstable: the standard deviation of pitch or roll over a 60 s (120-sample) moving window exceeds 2°.
- *Pressure* fails outside $[p_{\mathrm{ref}}/2,\ 2p_{\mathrm{ref}}]$, where p_{ref} is the median of the in-water samples ($p >$ 0.5 dbar) in the first 10% of the record (at least 1000 samples). A reference taken from the start of the record survives a sensor that saturates later in the deployment, which a whole-record median does not.
- *Temperature* fails outside the site's plausible range `cfg.qcOpts.Tvalid`, set to 9–26 °C in every catalog config (the code default, −2–40 °C, only catches gross failure). Where the thermistor is valid and the instrument's sound speed differs from the one implied by the measured temperature by more than 0.2%, velocity is rescaled and the segment is marked suspect downstream.

Failed samples are set to NaN in their own channel only.

Changed 2026-10. The April draft described a fixed ±5° pitch and roll limit, a whole-deployment median pressure reference, and row-level NaN-ing of every channel. The per-channel QC and the first-window pressure reference replaced those in July 2026 (`PIPELINE_NOTES.md`, `docs/OUTSTANDING_channel_decoupling.md`). A vertical-transit (off-bed) pressure test exists in `puv_channel_qc.m` but is off in every config.

Samples with a stable but non-zero tilt (for example from a pipe bent during a storm) are kept and rotated to the local vertical sample by sample, using the measured pitch ϕ and roll θ, whenever the median tilt exceeds 1°. The code applies roll first and then pitch,

$$\mathbf{v}_{\text{corrected}} = \mathbf{R}_{\text{pitch}}(\phi)\, \mathbf{R}_{\text{roll}}(\theta)\, \mathbf{v}_{\text{raw}}, \tag{1}$$

where

$$\mathbf{R}_{\text{roll}} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos\theta & -\sin\theta \\ 0 & \sin\theta & \cos\theta \end{bmatrix}, \qquad \mathbf{R}_{\text{pitch}} = \begin{bmatrix} \cos\phi & 0 & \sin\phi \\ 0 & 1 & 0 \\ -\sin\phi & 0 & \cos\phi \end{bmatrix}.$$

Corrected 2026-10. Earlier versions of this section, and the comment above the loop in `PUV_raw_process.m`, write the product as $\mathbf{R}_{\text{roll}}\mathbf{R}_{\text{pitch}}$. The expanded components in that loop are those of $\mathbf{R}_{\text{pitch}}\mathbf{R}_{\text{roll}}$, which (1) now states. Whether that order matches Nortek's pitch and roll convention has not been checked here.

The tilt-corrected velocity is rotated to a geographic buoy frame ($+x$ west, $+y$ north, $+z$ up, the left-handed frame used by CDIP MOP). For records in instrument (XYZ) coordinates the rotation uses the heading, either the median of the onboard compass or, where the compass drifts or steps at constant tilt, the mount heading logged in the field, plus the magnetic declination from the International Geomagnetic Reference Field (`igrfmagm`, IGRF-14 where the installed release supports it, otherwise IGRF-13 capped at 2025.0) at the deployment position and start date. Records stored in Nortek ENU coordinates, which are magnetic, have the same declination added (`enu_to_buoy_frame.m`).

Changed 2026-10. The declination for ENU records was added on 2026-10-07 (about 11.07° at San Diego; SOL24, LPL24, LPL25A rebuilt). The rule for preferring the logged heading dates from the same day. Twelve records whose logged and compass headings differ by more than 10° keep the compass value and are listed in `shared/heading_disputed_records.m`; their directional products carry at least that uncertainty.

1.3 Level 2: Spectral analysis

1.3.1 Segmentation and preprocessing

The L1 time series is divided into non-overlapping 1 h segments of 7200 samples, starting at the first UTC top of the hour, which matches the hourly cadence of the CDIP MOP hindcast. Random Doppler error in a segment-mean velocity falls as $1/\sqrt{N}$, so the 7200-sample mean has a noise floor $\sqrt{7200/2048} \approx 1.87$ times lower than the 2048-sample (17-minute) segments used before May 2026. Velocity and pressure products are validated separately: a segment's velocity products are kept when no more than 10% of its velocity samples are NaN (`segValid_vel`), and likewise for pressure (`segValid_p`). Remaining gaps are filled by linear interpolation in the interior and by holding the nearest valid value at a segment edge. Velocities are then rotated to a shore-normal frame ($+x$ onshore, $+y$ alongshore north) using either a shore-normal angle set in the config (`instr.shorenormal`) or the angle CDIP publishes for the MOP station (`instr.mopStation`).

Changed 2026-10. Before 2026-10-09 an edge gap was filled by extrapolating the interior trend, which inflated infragravity and velocity statistics on segments with edge gaps (Hs moved by at most 3.5 cm except one segment, U_b and u_{rms} by about 4% median and 25% at the 95th percentile on edge-gap segments; `docs/catalog_changes/2026-10-13_combined_rebuild.md`). Pressure-only and velocity-only validity replaced a single all-channel test in July 2026. The April draft also cited a within-hour stationarity audit; that audit is not reproduced in this repository.

For each valid segment the mean pressure gives the total depth

$$H = \frac{\overline{P} \times 10^4}{\rho g} + z_s, \tag{2}$$

with $\overline{P}$ the segment-mean pressure in dbar, $\rho = 1025\,\mathrm{kg\,m^{-3}}$ and $g = 9.81\,\mathrm{m\,s^{-2}}$. Segment means of u, v, w and temperature are stored before detrending. Pressure is converted to meters of water and all channels are linearly detrended.

1.3.2 Spectral estimation

Spectra are estimated with the multitaper method of Thomson (1982) (`shared/psd_multitaper.m`) on the whole 7200-sample segment, with time–bandwidth product $NW = 4$. The code keeps the Slepian tapers whose concentration eigenvalue exceeds 0.99, which for $NW = 4$ is $K = 6$ (the seventh has eigenvalue 0.937), giving about $2K = 12$ degrees of freedom per frequency bin. Bins are spaced $\Delta f = f_s/N = 2.78 \times 10^{-4}\,\mathrm{Hz}$; the resolution bandwidth is $2NW/T = 2.2 \times 10^{-3}\,\mathrm{Hz}$ for $T = 3600\,\mathrm{s}$, so neighboring bins are not independent. Auto- and cross-spectra among p, u and v use the same tapers, which keeps the directional moments consistent. For comparison, the Welch estimator used before April 2026 (Hann window, `nfft` = 256) had bins spaced $7.8 \times 10^{-3}\,\mathrm{Hz}$.

> Corrected 2026-10. Earlier versions of this document, the multitaper note and several repository docs state $K = 7$ tapers and 14 degrees of freedom. `psd_multitaper.m` has used the eigenvalue > 0.99 rule since it was written (April 2026), which for $NW = 4$ keeps six tapers (eigenvalues from `dpss(7200,4)`: 1.000, 1.000, 1.000, 1.000, 0.999, 0.993, 0.937, 0.699). The April draft also described the resolution of the multitaper estimate as comparable to Welch with 256-sample sub-segments; at a 1-hour segment it is finer by a factor of about 3.5 in bin spacing.

1.3.3 Pressure correction

The pressure spectrum is converted to a surface-elevation spectrum with the linear transfer function

$$K_p(f) = \frac{\cosh(kz_s)}{\cosh(kH)}, \qquad S_\eta(f) = \frac{S_p(f)}{K_p^2(f)}, \tag{3}$$

where $k(f)$ solves $\omega^2 = gk\tanh(kH)$ by Newton iteration (`shared/get_wavenumber.m`). Bins with $K_p < 0.1$ are set to zero to avoid amplifying noise at high frequency, and the resulting cutoff frequency f_{cut} is stored per segment.

1.3.4 Bulk wave parameters

Significant wave height $H_s = 4\sqrt{m_0}$, with $m_0 = \int S_\eta\,df$, is reported over the total band (0.004–0.25 Hz), the sea–swell band (0.04–0.25 Hz) and the infragravity band (0.004–0.04 Hz). The peak period $T_p = 1/f_p$ is the frequency of maximum S_η within the sea–swell band (argmax, not a centroid). The mean period $T_{m02} = \sqrt{m_0/m_2}$ and the energy flux $F = \rho g \int c_g S_\eta\,df$, with c_g the linear group velocity, are both evaluated over the total band. The mean wave direction is $\mathrm{atan2}(\bar{b}_1, \bar{a}_1)$, where $\bar{a}_1$ and $\bar{b}_1$ are the first-harmonic directional moments from the pressure–velocity cross-spectra, weighted by S_η over the sea–swell band (`shared/compute_bulk_params.m`).

1.3.5 Near-bed velocity and stress

Near-bed orbital velocity is obtained by scaling each frequency bin of the velocity FFT from the sensor height to the bed. Under linear theory $u(z) \propto \cosh(kz)/\sinh(kH)$ with z measured up from the bed, so the bed-to-sensor ratio is

$$\frac{u_{\mathrm{bed}}}{u_{\mathrm{sensor}}} = \frac{1}{\cosh(kz_s)}, \tag{4}$$

and the scaled spectrum is inverse transformed (`shared/bed_velocity_ifft.m`). U_b is the root-mean-square of $\sqrt{u_{\mathrm{bed}}^2 + v_{\mathrm{bed}}^2}$ over the segment, and A_w is the corresponding orbital excursion.

> Corrected 2026-10. Earlier versions of this section gave the factor as $\cosh(kH)/\cosh[k(H - z_s)]$, which is the form the code's own header records as the original, wrong one. Because $kz_s \approx 0.04$ for swell at 9 m, the factor in (4) differs from 1 by about 0.1% there.

Wave bed shear stress uses the Swart (1974) friction factor with T_p as the wave period. The Nikuradse roughness is $k_s = 2.5D_{84}$ where a measured grain-size distribution exists for the site (`shared/site_grain_size.m`); otherwise the legacy form $k_s = 10D_{50}$ is used with the placeholder $D_{50} = 0.25$ mm. As of catalog 2026-10-14.1, 19 of 65 records use a measured distribution, 4 an assumed site default of 217.4 µm and 42 the placeholder. Reynolds stresses ($\overline{u'w'}$, $\overline{v'w'}$, $\overline{u'v'}$) and turbulent kinetic energy come from the detrended velocities. Velocity skewness S_k and asymmetry A_s (Hilbert form, negative for the forward-pitched shoaling wave shape) follow Elgar and Guza (1985) and Hoefel and Elgar (2003).

> Corrected 2026-10. The April draft gave $k_s = 2.5D_{50}$ with a single 0.25 mm grain size. The grain-size tally is from the L3 provenance sidecars, as quoted in `README.md`.

1.4 Validation against the MOP wave model

> Historical method (April 2026). This subsection describes the comparison behind the results in Section 2. The scripts that implemented it are in `validation/archive/` (`compare_PUV_MOP.m`, `run_PUV_MOP_validation.m`). The fourth test below, peak broadening measured by Q_p on an interpolated grid, produced a result that was retracted on 2026-07-24 as a resolution artifact. The current comparison reduces both spectra to the model's frequency bins first (`validation/compare_shape_matched.m`, `shared/shoal_mop_to_site.m`).

PUV-derived wave parameters were compared with hourly output from the CDIP Monitoring and Prediction (MOP) system, a spectral wave model initialized from offshore CDIP buoys that provides $S_\eta(f)$ at the 10 m contour of each MOP transect. MOP spectra were shoaled to the PUV depth by conservation of energy flux,

$$S_\eta(f, h_{\mathrm{PUV}}) = S_\eta(f, 10\,\mathrm{m}) \times \frac{c_g(f, 10\,\mathrm{m})}{c_g(f, h_{\mathrm{PUV}})}. \tag{5}$$

Shoaled MOP wave heights were integrated over the sea–swell band with the non-uniform MOP bandwidth vector. In April 2026 the PUV segments were 17 min long and were interpolated to the MOP hourly grid. Agreement was summarized by R^2, RMSE, bias (PUV − MOP) and the Willmott (1981) skill score.

The frequency-resolved ratio $S_\eta^{\mathrm{PUV}}(f)/S_\eta^{\mathrm{MOP,shoaled}}(f)$ was averaged over matched hours in three bands: low swell (0.04–0.08 Hz), mid swell (0.08–0.14 Hz) and high swell (0.14–0.25 Hz).

Four candidate mechanisms for the systematic difference were tested: (1) nonlinear shoaling, through the correlation of the spectral ratio with the Ursell number $Ur = H_s L_p^2/h^3$; (2) directional narrowing during shoaling, by comparing the first-harmonic spread $\sigma_1 = \sqrt{2(1 - \sqrt{a_1^2 + b_1^2})}$ between PUV and MOP; (3) bound long waves, through the scaling of low-frequency excess with H_s^2; and (4) spectral peak broadening in the model, through the Goda peakedness $Q_p = (2/m_0^2) \int f\, S_\eta^2\, df$ and the half-power bandwidth.

1.5 Level 3: Wave forcing characterization

1.5.1 Frequency-band energy decomposition

The L2 surface-elevation spectrum is integrated over three bands, infragravity (0.004–0.04 Hz), swell (0.04–0.10 Hz) and sea (0.10–0.25 Hz), giving a band wave height, energy flux and mean period for each 1-hour segment, and a flag naming the band with the largest energy flux.

1.5.2 Storm event detection

Storm events are taken from the H_s series as exceedances of 1.5 m lasting at least 6 h, with events closer than 12 h merged. These thresholds are fixed in `L3_forcing/PUV_L3_storms.m`, not read from the config. To describe the forcing during PUV data gaps (for example when an instrument was damaged in a storm), hourly MOP output is loaded for a window extending 7 days beyond each end of the record, and storms found in the MOP record but absent from the PUV record are flagged as gap-period storms.

1.5.3 Transport proxies

Bottom energy flux $F_b = \rho g \int c_g S_\eta \, df$ is integrated over the sea–swell band and accumulated over the deployment for comparison with surveyed bed change. The Shields parameter $\theta = \tau_b / [(\rho_s - \rho) g D_{50}]$ uses the wave stress amplitude τ_b from L2, with the critical value from Soulsby (1997) (Soulsby–Whitehouse), and a segment is flagged as mobilized when θ exceeds it. Kinematic viscosity is computed per segment from the segment's mean temperature (`shared/seawater_nu.m`), and the settling velocity w_s follows Ferguson and Church (2004) with nominal-diameter constants $C_1 = 20$, $C_2 = 1.1$. The Rouse number $P = w_s / (\kappa u_*)$ uses the friction velocity of the mean combined wave–current stress τ_m (Soulsby, 1997), and is also given at D_{16}, D_{50} and D_{84} where a measured distribution exists.

Changed 2026-08 and 2026-10. Until 2026-08-13 the Rouse number used the oscillatory wave stress amplitude, which overstated suspension (median P 4.8 against a corrected 41.5 at TBR23 MOP580_5m, `docs/pipeline_levels.md`). Settling velocity used smooth-sphere constants ($C_2 = 0.4$) until 2026-08-13 and $C_2 = 1.0$ until 2026-10-09; viscosity was a fixed 1×10^{-6} m^2 s^{-1} until 2026-10-09. Six settling-velocity formulas could not be distinguished against the TBR23 surveys; they spread about ±15% at 250 µm.

1.5.4 Tidal current decomposition

Tidal harmonic analysis is run on the segment-mean velocities with `t_tide` (Pawlowicz et al., 2002), using complex input $u + iv$ so the cross-shore and alongshore constituents are fit together, and only over the span where velocity exists. Each call is seeded (`rng(0,'twister')`) because the bootstrap error estimate decides which constituents enter the prediction. The tidal depth reference is the NOAA prediction for La Jolla, Scripps Pier (station 9410230), fetched through the lab toolbox function `getztide2`; in the catalog build every record has NOAA depth. Subtidal residual currents (observed minus tidal) carry the wave-driven mean flows, including undertow and longshore currents.

For TBR23 the NOAA prediction matched the PUV depth record better than a `t_tide` fit to the PUV pressure ($R = 0.995$ against 0.987), and the lag of maximum correlation was 0 h, which confirmed that the TBR23 instrument clocks were in UTC. The same test later showed that the file-name clocks of the pre-2019 recorders were Pacific local time (see Level 1); `scripts/audit_clock_lag.m` repeats it for those records.

Changed 2026-10. The seeding and the velocity-span restriction date from 2026-10-09. A change set in progress on 2026-10-15 makes the tide station a per-deployment setting and makes a missing `t_tide` fatal; until it lands, every site uses station 9410230, which is the wrong tide for a site far from La Jolla. See `CHANGELOG.md` for the current behavior. The R values for TBR23 are from the April analysis and were not recomputed.

1.6 Mean cross-shore flow validation

Historical method (May 2026). This is the test design behind the mean-flow results in the Results section. It ran on 38 instruments at both 1-hour and 17-minute segmentation. The 17-minute archive was deleted from the server on 2026-07-28, so the segmentation comparison cannot be rerun until a 17-minute L2 is regenerated from the current L1 (`validation/archive/compare_seglen_phase2.m`).

The segment-mean cross-shore velocity $\overline{u}$ was tested against three hypotheses (wave-driven Stokes return flow, a constant per-instrument calibration offset, and random Doppler noise) on every PUV deployment then in the catalog (38 instruments at Torrey Pines, Los Peñasquitos lagoon, Solana Beach, SIO Pier, Catalina Island, and the Ruby2D cross-shore array).

1.6.1 Linear scaling against H_s^2

For each instrument the segment-mean cross-shore velocity was fit as

$$\overline{u} = \alpha\, H_s^2 + \beta + \varepsilon, \tag{6}$$

restricted to segments with $H_s \geq 0.2\,\mathrm{m}$. Stokes return-flow theory (Longuet-Higgins, 1953) predicts a depth-averaged Eulerian compensation flow $\overline{u} \approx -gH_s^2/(16ch)$ for shallow water with $c = \sqrt{gh}$, so $\alpha_{\mathrm{theory}} = -g/(16ch)$, about -0.018 at $h = 5\,\mathrm{m}$ and -0.003 at $h = 15\,\mathrm{m}$ in $\mathrm{m\,s^{-1}}$ per $\mathrm{m^2}$ of H_s^2, with $\beta_{\mathrm{theory}} = 0$. The hypotheses predict different coefficients. Real return flow gives α matching theory in sign and within a factor of order one, with β centered on zero across instruments. A constant offset gives $\alpha \approx 0$ with β scattered at the offset magnitude. Random noise leaves both coefficients unconstrained but shrinking toward zero with longer averaging.

1.6.2 Three robustness checks

First, the same fit was run on the alongshore component $\overline{v}$, to test whether α is specific to cross-shore physics or reflects storm-coherent forcing (wind stress, atmospheric setup, alongshore gradients) that also correlates with H_s. Second, the fit was restricted to $H_s \geq 1\,\mathrm{m}$ and the change in R^2 reported. Third, the alongshore model was extended to

$$\overline{v} = \alpha_{v0}\, H_s^2 + \alpha_{v1}\, \sin(2\theta_{\mathrm{rel}})\, H_s^2 + \beta_v, \tag{7}$$

where θ_{rel} is the wave incidence angle relative to shore normal, taken from the L2 mean direction in the rotated frame. An error θ_{err} in the shore-normal angle predicts $\alpha_{v0} = \alpha_u \tan\theta_{\mathrm{err}}$ with $\alpha_{v1} \approx 0$, while a radiation-stress-driven longshore current predicts a dominant α_{v1} with the sign of the local wave climate. The implied $\theta_{\mathrm{err}} = \arctan(\alpha_{v0}/\alpha_u)$ is an upper bound on any rotation error.

1.6.3 Auxiliary diagnostics

(1) Tidal modulation of $\overline{u}$ was measured by binning segment means into 16 phase bins of band-passed depth (0.0019–0.0093 Hz, periods of about 30 min to 3 h), stratified by H_s; a constant offset has zero modulation in every stratum. (2) Time series of $\overline{u}$ from independently calibrated instruments were

correlated at common timestamps. (3) The Reynolds stress $\overline{u'w'}$, a segment mean of fluctuation products rather than of the velocity itself, was correlated with $\overline{u}$; independent noise in either path would push that correlation toward zero.

1.6.4 Segmentation independence

The validation was run at 1-hour and 17-minute segmentation. Random noise predicts that $\overline{u}$ estimates shrink toward zero as the averaging window grows. Real wave-driven flow predicts α roughly independent of segmentation, since it is a kinematic relation, with $|\beta|$ tightening slightly toward zero.

2 Results

Historical results, not recomputed. Everything in this section was computed between April and July 2026 on the catalog as it then stood: 33 to 38 instruments for most subsections, 17-minute segments for the earliest (bulk agreement, spectral bias, Ruby2D, L3, TBR23 recovery), 61 records for the July hypothesis tests. Since then the catalog has been rebuilt several times: heading corrections (2026-05-14, 2026-07-27, 2026-10-07), `doffp`, coordinate and clock corrections (2026-07-27/28), the edge-gap fill fix and new sediment physics (2026-10-09), and the full L2 to L4 rebuild for catalog 2026-10-14.1. Treat every number below as a record of what was found at the time. Current versions of the model-comparison results are maintained in the `Wave_model_validation_paper` repository, where a reconciliation against catalog 2026-10-14.1 was in progress on 2026-10-15.

2.1 PUV–MOP validation

2.1.1 Bulk wave parameter agreement

Of 40 instruments across 20 deployments, 33 were processed successfully (82.5%), with 7 failures attributable to hardware issues (burial, bent pipes, dead batteries). PUV-derived wave parameters showed strong agreement with the MOP wave model across all four TBR23 instruments (the summary table planned here was never drafted). For instruments unaffected by biofouling, H_s correlations exceeded $R^2 = 0.83$ with RMSE < 0.06 m and biases of 0.002–0.008 m. One instrument (MOP580 5m) exhibited lower agreement ($R^2 = 0.66$, bias $= 0.037$ m), attributed to intermittent kelp fouling that reduced the valid segment rate to 49% compared with 85–91 % for the remaining instruments. A small but consistent positive H_s bias was observed across all instruments, with PUV-measured wave heights systematically exceeding shoaled MOP values.

2.1.2 Frequency-dependent spectral bias

The frequency-resolved spectral ratio $S_\eta^{\mathrm{PUV}}(f)/S_\eta^{\mathrm{MOP,shoaled}}(f)$, averaged across all matched hours, revealed that the positive H_s bias was concentrated at the swell spectral peak (0.04–0.08 Hz), where the pressure transfer function $K_p > 0.9$ and the correction is minimal (table planned, never drafted). The mid-swell band (0.08–0.14 Hz) showed near-unity ratios (0.99–1.03), and the high-swell band (0.14–0.25 Hz), where the pressure correction is most aggressive (K_p as low as 0.24), also showed only modest excess (1.03–1.05). The low-swell amplification increased with decreasing depth, from a band-averaged ratio of 1.17 at 8.3 m to 1.28 at 5.5 m. The overall excess energy relative to the shoaled MOP spectrum was 4.5 % at 8.3 m depth and 8.7 % at 5.5 m.

The concentration of excess energy at the spectral peak, where the pressure correction factor is close to unity, indicates that the bias does not originate from the pressure-to-surface-elevation transfer function.

Superseded 2026-07. The apparent concentration of excess at the peak came from comparing spectra on mismatched frequency grids (see the retraction under "MOP spectral peak broadening" below). On the model's own frequency bins the excess sits near the second harmonic, $f/f_p \approx 2$.

2.1.3 Hypothesis testing

Rewritten 2026-07-27; values partly superseded 2026-07-28. This subsection replaced an April version that ruled out the first three mechanisms in favor of the fourth. The values are from `validation/redo_hypothesis_elimination.m` as run on 2026-07-27 (72,948 hours, 61 records). After the 2026-07-28 catalog refresh, `Wave_model_validation_paper/docs/findings_refresh_2026-07-28.md` records these values as moved (not recomputed here): $\rho(Ur, \nu)$ $+0.332 \rightarrow +0.347$; ν in the most linear H_s/h bin $1.000 \rightarrow 1.003$; the ratio at $f/f_p = 2.1$ for $H_s/h > 0.12$, $1.338 \rightarrow 1.332$; bicoherence ordering $-0.008 \rightarrow +0.017$ (lowest band) and $+0.285 \rightarrow +0.304$ (highest); σ_1 ratio $1.156 \rightarrow 1.157$; matched-grid Q_p ratio $1.008 \rightarrow 1.009$. The verdicts did not change on that refresh. None of these has been recomputed on catalog 2026-10-14.1.

Four candidate mechanisms for the systematic positive bias were evaluated (summary table planned, never drafted).

Nonlinear shoaling. Triad interactions during shoaling transfer energy from the spectral peak to its super-harmonic near $2f_p$. The PUV/model energy ratio, formed on the model's own frequency bins and binned by f/f_p, shows that transfer directly: a trough of 0.94–0.95 at $f/f_p = 1.3$–1.5 and a maximum of 1.34 at $f/f_p = 2.1$ for hours with $H_s/h > 0.12$, against 1.10 for hours below it. The discrepancy is organized by the Ursell number ($\rho = +0.332$, $N = 72\,948$ hours across 61 records) rather than by depth, which drops to $\rho = -0.114$ once H_s/h is held fixed. Binned by H_s/h, the spectral-width ratio $\nu^{\mathrm{PUV}}/\nu^{\mathrm{MOP}}$ rises monotonically from 1.000 below $H_s/h = 0.04$ to 1.100 above 0.20: the model and the observations agree to three decimal places when the waves are linear, and diverge as they cease to be. Bicoherence at the swell self-interaction orders the harmonic excess *within* fixed H_s/h bands, from $\rho = -0.008$ below $H_s/h = 0.06$ to $\rho = +0.285$ above 0.20: flat where triad coupling cannot operate, strengthening where it can.

An earlier version of this analysis correlated the Ursell number against the *low-swell* band ratio (0.04–0.08 Hz) and found $R^2 = 0.004$. That test cannot detect the mechanism: nonlinear shoaling deposits energy near $2f_p \approx 0.18$ Hz, not at the peak.

Directional narrowing. The first-harmonic directional spread σ_1, from the PUV cross-spectral a_1 and b_1, is *wider* than the MOP spread at swell frequencies, by a median factor of 1.156 across 61 records ($p = 2.6 \times 10^{-6}$). The sign is opposite to the narrowing expected from refraction-driven focusing, and the wider PUV spread is consistent with scattered energy at shallow depths that the forward-propagating model does not carry. The spread ratio does not covary with the discrepancy it was proposed to explain ($p = 0.30$ against the H_s ratio, $p = 0.54$ against ν), so it is a separate bias in the model's directional distribution rather than the mechanism behind the energy difference. Because σ_1 depends on $r_1 = \sqrt{a_1^2 + b_1^2}$, it is invariant to instrument rotation and unaffected by the heading corrections applied in 2026-07.

Bound long waves. Second-order difference-frequency forcing predicts bound infragravity variance from the swell band via the Hasselmann (1962) / Schäffer–Madsen (1995) kernel. The ratio of predicted-bound to observed-total IG variance rises with H_s/h ($\rho = +0.913$ across 61 record medians; +0.758 per hour) and crosses unity near $H_s/h = 0.12$, above which second-order theory over-predicts the bound component. Against the spectral-width discrepancy the same ratio gives $\rho = +0.288$, falling to +0.149 once H_s/h is controlled. Bound-wave forcing and super-harmonic generation both arise from triad interactions among the same swell components, so this collinearity is expected; the two are aspects of one nonlinear response rather than competing explanations, and the data here do not separate them.

An earlier version rejected this hypothesis partly on the grounds that the excess energy sat at the swell peak rather than at group frequencies. That apparent concentration at the peak was itself the grid artifact described below.

Retracted 2026-07-24. The result in the next two paragraphs is an artifact of the comparison method, not a property of the MOP model. It was computed after interpolating the model's roughly 20 frequency bins onto the PUV grid (`validation/archive/analyze_spectral_shape.m`), which manufactures a broad, smooth peak and lowers $\int S^2\,df$. Comparing a PUV spectrum with a degraded copy of itself through the same path gives a Q_p ratio of 1.225 and 68% apparent narrowing (TBR23/MOP586_7m, 1382 segments), more than the whole effect reported here. On a resolution-matched grid the effect is absent; see "Cross-deployment confirmation" below. The paragraphs are kept as the record of what was claimed. Quote nothing from them.

MOP spectral peak broadening. The Goda peakedness parameter Q_p was systematically higher for PUV spectra than for shoaled MOP spectra (median $Q_p = 1.88$ vs. 1.75, ratio 1.07). The PUV half-power spectral bandwidth was 16.7 % narrower than MOP (0.039 Hz vs. 0.047 Hz), and the PUV peak spectral density was 14.4 % higher. The peakedness difference $\Delta Q_p = Q_p^{\mathrm{PUV}} - Q_p^{\mathrm{MOP}}$ showed the strongest correlation with H_s amplification of any tested mechanism ($R = 0.44$, $R^2 = 0.20$). Hours when the PUV spectrum was more peaked relative to MOP corresponded to hours with larger positive H_s bias.

These results indicate that the MOP spectral wave model introduces numerical broadening of the spectral peak during propagation from the initialization depth (10 m) to the instrument depth. The in-situ PUV measurement captures the true, narrower spectral peak shape, resulting in higher peak energy density and thus higher H_s for a given total variance. This spectral broadening is a recognized property of discretized spectral wave models (cited in the April draft by key only: `rogers2002`, `booij1999`) and has not been previously quantified using co-located PUV observations in this depth range.

2.1.4 Cross-deployment confirmation

Rewritten after the 2026-07-24 retraction. The April version of this subsection reported the broadening pattern as reproduced across 11 instruments.

On a resolution-matched grid the peak-shape difference is absent across the catalog: the median Q_p ratio is 1.008 over 62 instrument-records ($p = 0.44$), with an interquartile range of 0.980–1.039 that spans unity. The legacy interpolated-grid calculation returns 1.203 on the same records, within round-off of the 1.225 obtained by comparing a PUV spectrum against a degraded copy of itself.

Consistency of the original pattern across instruments was read as evidence that it was physical. It is not: the artifact is a property of the comparison, so it appears in every record by construction, and agreement across deployments at matched depths is what a shared upstream error produces.

2.1.5 Pipeline cross-validation: Ruby2D head-to-head

April 2026, 17-minute segments. The new-pipeline side of this comparison used 2048-sample segments ($\Delta f = 9.77 \times 10^{-4}$ Hz) and the L2 code of that date. The later, isolated test of each legacy fault (`docs/legacy_pipeline_defects.pdf`, 3,713 segments, 10 records) finds the band-averaged spread error reaching 17° at the 95th percentile and the b_2 moment, which radiation stress is built from, unreliable. The 1° median spread offset reported here is consistent with that, but the conclusion that prior Ruby2D statistics need no revision applies to H_s, T_p and mean direction only. The legacy source as it now stands sets `nfft` = 2400 for both auto- and cross-spectra, with comments recording a change from 7200. The archived Ruby2D L2 reports T_p at 1200, 1800 and 3600 s, the first three bins of a 2.78×10^{-4} Hz grid, so it was built before that edit.

To verify that the multi-taper L2 implementation does not introduce biases relative to an established processing pipeline, we reprocessed the Ruby2D MOP582_6m record (Torrey Pines, October 2021 to February 2022, ∼3,100 hours) and compared it segment-by-segment against the archived L2 product produced by an independent prior pipeline (`beachpeeps/PUV_Processing`). The prior pipeline uses a single-periodogram estimator (Hann window, full 60-minute segments, $\Delta f = 2.78 \times 10^{-4}$ Hz) and applies a fixed cap of $K_p^2 \leq 10$ to the pressure transfer function. The legacy record has 3,095 60-min segments; the new pipeline produces 10,903 17-min segments over the same record. Segments were matched on midpoint nearest neighbor with a tolerance of 30 minutes, yielding 2,322 matched pairs (75%). The remaining 25% are hours that the new pipeline rejects under the segment-validity criterion (`nanMaxFrac` = 0.10) but that the prior pipeline retains by zero-padding small NaN gaps.

Table 1: Ruby2D MOP582_6m: bulk wave parameter agreement between the new multi-taper pipeline and the legacy archived L2. T_p statistics are computed after restricting both estimates to the SS band (4–25 s); without this filter, three segments on January 15, 2022 in the prior pipeline report T_p at the IG/DC bin (T_p = 1200, 1800, 3600 s) because the prior pipeline's peak picker is not band-limited.

Metric	Legacy (med.)	Multi-taper (med.)	Bias (M−L)	RMS	R^2
H_s (m)	0.771	0.757	−0.009	0.053	0.981
T_p (s)	13.48	13.30	−0.02	1.48	0.761
Direction SS (°)	−7.3	−7.2	+0.2	1.2	0.927
Spread SS (°)	14.7	15.8	+1.0	1.7	0.877

Bulk wave parameter agreement is excellent across all four metrics (Table 1; Figure 1). H_s agrees to 5 cm RMS with a −9 mm median bias ($R^2 = 0.98$), and direction agrees to within 1.2° RMS ($R^2 = 0.93$). The directional spread is 1° wider in the new pipeline on average; this is the largest systematic offset observed and remains within the segment-to-segment scatter. The new T_p shows the discrete quantization bands expected from $\Delta f = 0.000976$ Hz versus the prior pipeline's 0.000278 Hz, but no systematic bias.

Per-segment surface elevation spectra agree closely in shape (Figure 2). At a representative median segment ($H_s = 0.77$ m, November 27, 2021) and at a storm peak ($H_s = 2.78$ m, December 14, 2021, bimodal sea state), the integrated H_s values from the two pipelines agree to within 4%. The visible difference between the spectra is the chi-square noise characteristic of a single-periodogram estimator in the prior pipeline, versus the variance reduction provided by averaging $K = 7$ DPSS tapers in the new pipeline. The underlying peak shapes and locations agree.

This cross-validation establishes two things. First, the multi-taper implementation reproduces an independent prior pipeline to within instrument noise on bulk wave parameters, confirming that the estimator change is a refinement rather than a methodological break. Prior published Ruby2D bulk wave statistics do not require revision. Second, the consistency-of-windows fix in the new cross-spectral calculations (auto- and cross-spectra both use the same DPSS tapers, unlike the prior pipeline which mixed Hanning and Hamming) does not produce the order-20% directional errors that the bug analysis suggested were possible at this site: the median impact at Ruby2D MOP582_6m is approximately 1° in spread.

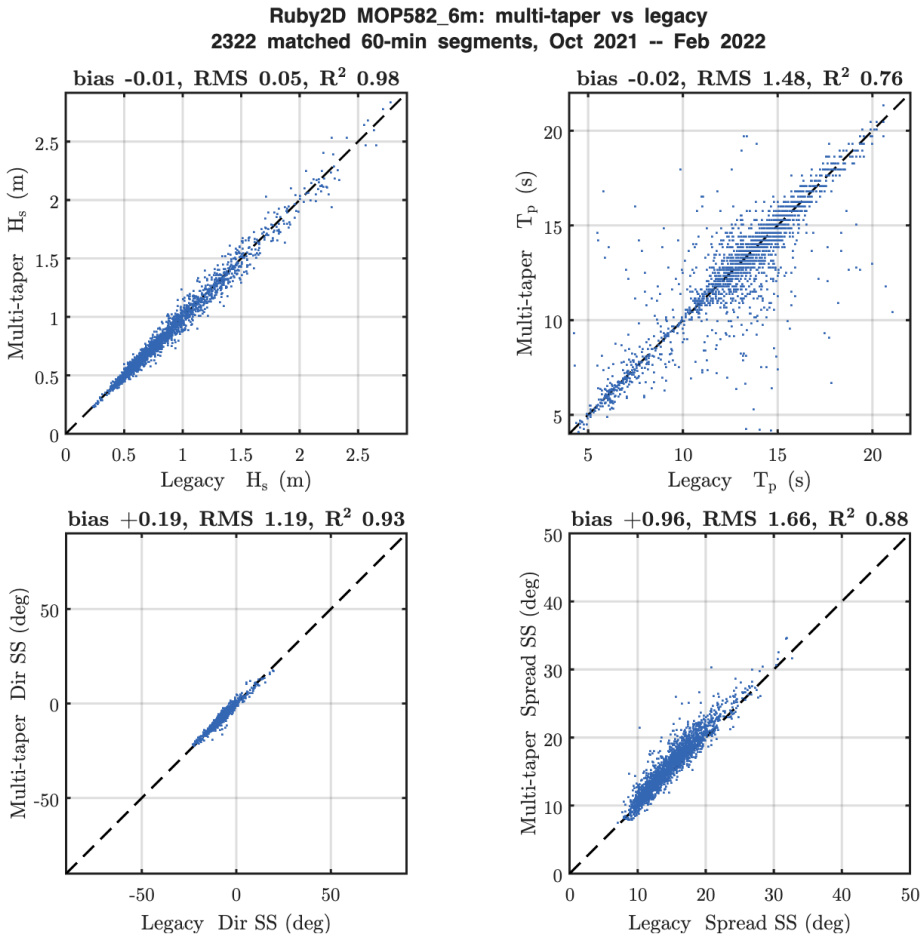


Figure 1: Bulk parameter agreement between the new multi-taper pipeline (vertical axis) and the legacy archived L2 (horizontal axis) for Ruby2D MOP582_6m, 2,322 matched 60-minute segments, October 2021 to February 2022. The dashed line is 1:1.

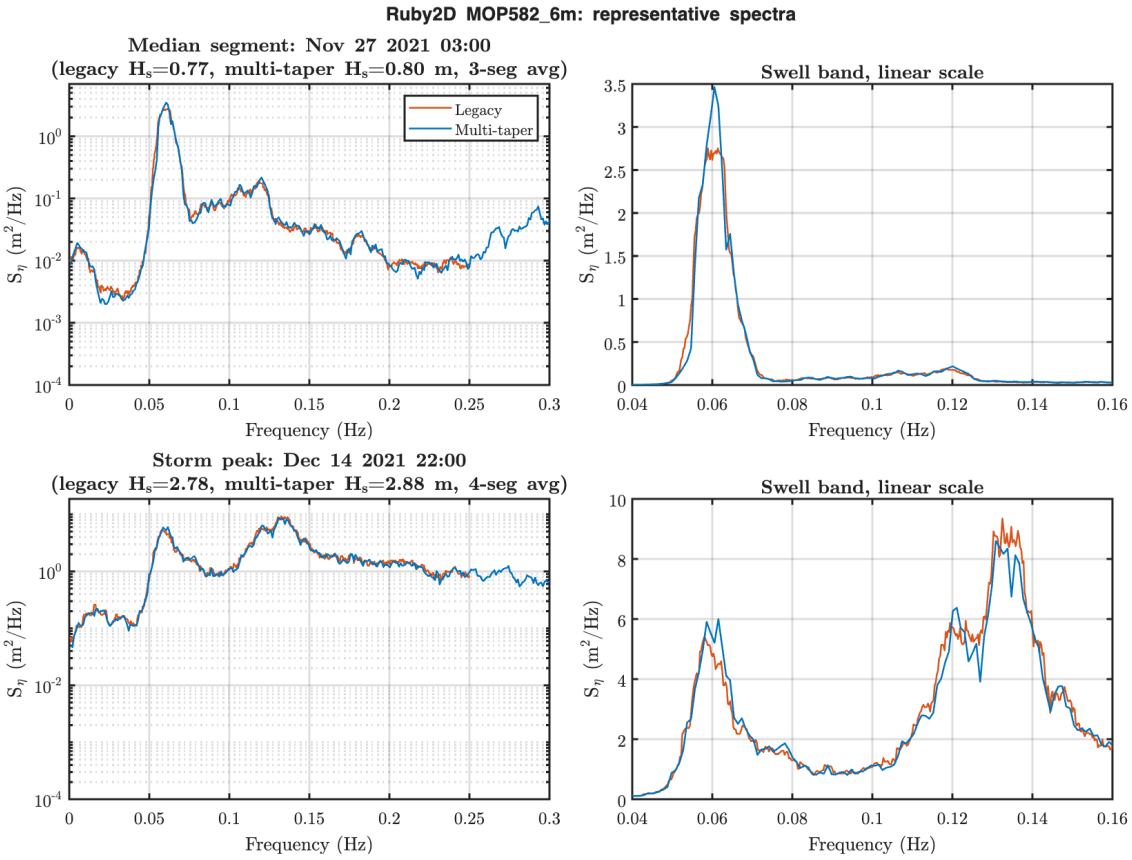


Figure 2: Representative surface elevation spectra for Ruby2D MOP582_6m. Top: median segment ($H_s = 0.77$ m). Bottom: storm peak ($H_s = 2.78$ m). Left column: log scale, full SS band. Right column: linear scale, swell band detail. The multi-taper estimates (blue) are smoother than the prior single-periodogram estimates (red); peak locations and total variance agree.

2.1.6 Site-specific amplification at SIO Pier

The SIO Pier instruments (MOP 511, 7m depth) exhibited dramatically larger PUV-to-MOP spectral amplification than all other sites. The peak spectral density ratio ranged from 2.0 to 2.7 across four deployment periods (SIO24A through SIO25E), with corresponding H_s amplification of 29–36%, far exceeding the 1–11% observed at Torrey Pines, Solana Beach, and Los Peñasquitos sites at comparable depths.

Investigation confirmed that the MOP frequency grid is identical across all stations (20 bins, 0.04–0.40 Hz), ruling out a resolution artifact. The MOP model total energy at 10 m depth is comparable between SIO (H_s = 0.62 m) and Torrey Pines (H_s = 0.61 m), yet the in-situ PUV at SIO measures approximately 2.5× the peak spectral density predicted by the shoaled MOP spectrum.

The SIO Pier PUV is located near the head of Scripps Submarine Canyon, where canyon-edge refraction is known to create wave energy focusing (the April draft cites these by bibliography key only, `munk1948` and `shepard1950`; the references were not identified or checked in the 2026-10 revision). The canyon bifurcates nearshore, and wave focusing at the canyon head has been documented at Black's Beach on the adjacent fork. The MOP model, which propagates spectra along a relatively coarse alongshore grid, does not resolve this fine-scale bathymetric focusing. The PUV measurements capture the true, focused wave field, resulting in the observed amplification.

> The comparison with "universal spectral peak broadening" in the next paragraph refers to the result retracted on 2026-07-24. The SIO amplification itself was measured on the then-current spectra; whether it survives the resolution-matched comparison was not checked in this document.

This site-specific effect is distinct from the universal spectral peak broadening described above. While broadening produces a modest (∼5–10%) Hs bias that is consistent across all sites, the canyon focusing at SIO produces an order-of-magnitude larger amplification that is unique to that location.

2.2 L3 wave forcing characterization

> April 2026, 33 instruments, 17-minute segments. The Rouse numbers below were computed from the oscillatory wave-stress amplitude, which overstated suspension; since 2026-08-13 the Rouse number uses the mean combined wave–current stress, and the corrected values are larger (median 41.5 against 4.8 at TBR23 MOP580_5m). The bedload conclusion is unchanged in sign, but the quoted range is obsolete. Shields values used D_{50} = 0.25 mm everywhere; TBR23 now uses measured grain sizes. The "UTC clock convention" was confirmed for the instruments tested here; the pre-2019 file-name clocks were later found to be Pacific local time.

2.2.1 Energy band decomposition

Frequency-band analysis revealed clear seasonal patterns across all deployments. Winter deployments (TOR23W, SOL23, TOR24W, SOL24) were swell-dominated (65–90% of total energy flux in the 0.04–0.10 Hz band), while summer deployments (TBR23, SIO25B–D) showed approximately equal swell and sea contributions. The energy flux consistency check confirmed exact agreement between the sum of band fluxes and the total L2 energy flux ($F_{\text{total}}/F_{\text{L2}}$ = 1.0000) for all 33 instruments.

2.2.2 Storm characterization with MOP gap-filling

Storm detection identified 0 events during the calm summer TBR23 deployment and up to 11 events during winter TOR23W, including the major December 28, 2023 storm sequence (H_s up to 3.6 m

at 7 m depth, sustained over 74 hours). MOP gap-filling proved essential: the SOL23 MOP651 7m instrument detected 0 storms from PUV data (55% valid segments) but MOP identified 3 events during data gaps, including an 86-hour storm with $H_s = 3.09$ m. Across all deployments, MOP supplemented 1–8 additional storm events per instrument that were missed due to PUV data gaps during the most energetic conditions.

2.2.3 Transport proxy patterns

The Shields parameter showed clear depth and seasonal dependence. Median Shields numbers decreased from 0.24 at 5.6 m to 0.065 at 15.5 m depth (TOR23W MOP586 cross-shore array), with corresponding mobilization fractions decreasing from 100% to 78%. Winter deployments showed approximately 3× the cumulative bottom energy flux of summer deployments at equivalent depths. The Rouse number indicated bedload-dominated transport (median $P = 2.8$–6.0) across all sites and seasons, with the lowest values (most suspension-capable) at the shallowest winter sites.

2.2.4 Tidal and subtidal currents

Tidal currents were small (typically <0.05 m/s) relative to wave-driven subtidal flows. The subtidal cross-shore residual was consistently negative (offshore-directed) across all sites and seasons, with magnitudes of 0.004–0.023 m/s, consistent with wave-driven undertow. Tidal depth predictions from the Scripps Pier NOAA gauge matched PUV depth anomalies at $R > 0.99$ for instruments with continuous records, confirming the UTC clock convention and validating the pressure-to-depth conversion.

2.3 TBR23 post-storm recovery dynamics

April 2026. This analysis moved to the Paper_1 repository, which holds the current numbers. Note for the MOP580_5m exclusion below: on 2026-05-14 that record's L1 heading was found to be 180° wrong (a field-sheet entry of 270.74° for 90.74°) and was corrected. A 180° heading error reverses the sign of u, which by itself produces negative velocity skewness and anti-correlation of $\overline{u}$ with neighboring instruments, the two symptoms attributed here to kelp fouling. The fouling attribution of those symptoms is therefore doubtful. The low valid-segment rate (49%) does not depend on heading and was not re-examined.

2.3.1 Observed morphological change

Jetski bathymetric surveys (22 surveys during the deployment period) revealed 25–39 cm of net erosion at the PUV instrument locations over the May–August 2023 deployment, with the 5 m sites eroding more than the 7 m sites. Pre-deployment context from the MOPS survey archive showed that the winter 2023 storm sequence had deposited approximately 85 cm of sand at the 5 m depth contour relative to the prior year, placing the bed well above its long-term equilibrium profile. The TBR23 deployment thus captured the post-storm readjustment phase rather than active storm erosion.

2.3.2 Competing transport mechanisms

Analysis of the PUV velocity moments revealed two competing cross-shore transport mechanisms operating simultaneously during the recovery period:

Onshore: velocity skewness. Velocity skewness S_k was positive at all three reliable instruments (median $S_k = +0.05$ to $+0.12$), indicating peaked crests and flat troughs characteristic of shoaling waves. Skewness increased with decreasing water depth, following the Ruessink et al. (2012) parameterization based on the Ursell number. The skewness-driven bedload flux $\langle u|u|^2 \rangle$ was directed onshore, representing the wave-driven recovery mechanism.

Offshore: undertow. The subtidal mean cross-shore current was consistently offshore-directed ($\overline{u} = -0.006$ to -0.012 m/s) at all three instruments, representing wave-driven return flow (undertow). Undertow magnitude increased with wave height and was modulated by tidal phase, with stronger offshore flow at high tide when more wave energy reached the instrument depths.

2.3.3 Depth-dependent transport balance

The ratio of onshore skewness flux to offshore undertow flux showed a clear depth dependence exploiting the tidal modulation of water depth at each fixed instrument location (∼2.5 m tidal range). At the shallowest site (MOP586 5m, median depth 5.5 m), the skewness flux exceeded the undertow flux by a factor of 3–4, indicating net onshore transport and active recovery. At the 7 m sites (median depth 8.3 m), the undertow flux slightly exceeded the skewness flux, indicating near-equilibrium or weak offshore transport.

The crossover depth where onshore and offshore fluxes balanced was approximately 6–7 m, defining a *recovery boundary* below which the post-storm deposit was actively being redistributed shoreward. This is consistent with the survey observations: the MOP586 5m site exhibited a 25 cm recovery pulse in mid-July 2023 that was not observed at the deeper instruments.

One instrument (MOP580 5m) was excluded from the transport analysis due to intermittent kelp fouling that produced anomalous negative velocity skewness (92% of segments) inconsistent with the other instruments and with wave theory.

2.4 Mean cross-shore flow validation

May 2026 (run of 2026-05-05). This run predates the 2026-05-14 heading corrections: TOR24S/MOP586_7m entered it with its velocity sign reversed, and TBR23/MOP580_5m was excluded as kelp-fouled for symptoms a reversed heading also produces (see the note on the TBR23 recovery section). It also predates the 2026-07 and 2026-10 heading changes and the 2026-10 edge-gap fill fix. The fractions of instruments with the predicted sign of α are the numbers most exposed. Not rerun since.

The validation framework defined in Section 1.6 was applied across the 2023+ catalog (33 instruments at four sites: 19 at Torrey Pines, 4 at Los Peñasquitos lagoon, 5 at Solana Beach, 5 at SIO Pier). The same fits were repeated at the legacy 17-minute segmentation to provide a segmentation-independence check; pre-2023 deployments (Catalina Island, Ruby2D Torrey cross-shore array; total 5 additional instruments) were added at 1-hour only. The analysis excludes the kelp-fouled TBR23 MOP580 5m record and the SIO25C partial deployment ($N = 61$ segments, $R^2 = 0.001$).

2.4.1 Cross-deployment headline

The results decisively favor the real-Stokes-return-flow hypothesis (headline table planned, never drafted; values are in `docs/mean_flow_validation_plan.md`, which is kept out of the public repository). The intercept distribution is centered on zero across both segmentations (median $|\beta| = 0.6\,\mathrm{mm\,s^{-1}}$ at 17-min, $2.8\,\mathrm{mm\,s^{-1}}$ at 1-hour), and 100 % of 17-min and 94 % of 1-hour intercepts fall inside a ±2 cm/s band that envelopes the manufacturer-quoted single-sample velocity accuracy at the deployed range setting. The slope distribution clusters near the Stokes prediction

($\alpha/\alpha_{\text{theory}} = +0.53$ at 17-min, +0.71 at 1-hour), with 66–81 % of instruments showing the predicted sign of $\alpha < 0$ (offshore-directed undertow). Going from 17-min to 1-hour averaging *strengthens* the slope-vs-theory match and the correct-sign fraction; random instrument noise would do the opposite.

2.4.2 Site decomposition

The Torrey Pines site (19 instruments across 5 deployments and depths from 4.8 to 15.5 m) gives the cleanest theory match: 83 % have the correct α sign and median $\alpha/\alpha_{\text{theory}} = +0.77$ at 17-min. The Los Peñasquitos lagoon site agrees (+0.88 ratio, 75 % correct sign at 17-min). At Solana Beach and SIO Pier, the 17-min slopes are mixed in sign with low R^2, but both sites *flip* to predominantly negative α when re-fit at 1-hour (Solana: 20 % → 100 % correct sign; SIO Pier: 40 % → 80 %). This is interpreted as longer averaging exposing a wave-driven signal that the 17-min sampling variance was partially obscuring; the noise hypothesis predicts the opposite shift.

2.4.3 Tidal-phase modulation

Across all instruments, the median peak-to-peak modulation amplitude of $\overline{u}$ at $H_s \geq 1\,\text{m}$ is $2.6\,\text{cm}\,\text{s}^{-1}$ (1-hour) and $2.8\,\text{cm}\,\text{s}^{-1}$ (17-min), with peaks reaching 6–8 $\text{cm}\,\text{s}^{-1}$ at multiple Torrey instruments. A constant instrumental offset has zero modulation amplitude in any H_s stratum by definition; the observed amplitudes are 30–80 times the segment-mean random-noise floor and several times larger than the largest fitted intercepts.

2.4.4 Cross-instrument and Reynolds-stress consistency

Time correlations of $\overline{u}(t)$ between independently- calibrated Vectors at common timestamps were positive across all non-fouled instrument pairs in TBR23, reaching $R = +0.57$ between the two 7-m records. The kelp-fouled MOP580 5m record showed *anti*-correlation with all peers ($R = -0.33$ to -0.48), providing an internal consistency check against the fouling-flag exclusion. The Reynolds stress component $\overline{u'w'}$, derived from the same record but via an independent processing path, correlated positively with $\overline{u}$ at all four TBR23 instruments ($R = +0.24$ to $+0.63$).

2.4.5 Wave-direction discrimination

The extended alongshore model (Eq. (7)) was decisive against the shore-normal-rotation-error hypothesis. Across 30 instruments at 1-hour segmentation, the wave-direction-dependent coefficient α_{v1} dominated the wave-direction-independent coefficient α_{v0} in magnitude in 90 % of records, with the 95 % confidence interval on α_{v1} excluding zero in 83 % of records. The sign of α_{v1} was positive in 80 % of cases, corresponding to NW swells driving southward longshore current and S swells driving northward longshore current, the classical radiation-stress signature for the WNW-facing San Diego County coast. Reading α_{v0}/α_u as an upper-bound implied shore-normal angle error places that bound at a few degrees, consistent with the precision of the CDIP MOP shore-normal metadata.

2.4.6 Verdict

The cross-deployment dataset is incompatible with both the constant-offset and the random-noise hypotheses, and matches the Stokes return-flow prediction in sign, magnitude, and segmentation-length response. Random Doppler noise reduces the segment-mean noise floor to ∼0.4 mm/s at

17-min and ∼0.24 mm/s at 1-hour averaging, well below any of the slopes, modulation amplitudes, or cross-instrument shared variance reported above. The small (mm/s-scale) wave-direction-independent residuals visible in the alongshore fit are consistent with site-specific Eulerian residual currents and any small remaining shore-normal-rotation error; they are not the dominant component of either velocity component's response to wave forcing.